Full Universe NSE Stock Screener

Technical Architecture & Scoring Methodology

Document type: Technical Reference Report
Version: 2.0 (Recommendation Strategy v3)
Universe: ~181 NSE tickers, 8 sector buckets
Stack: Python screener | FastAPI | React UI

Generated from quant screener codebase

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1. Executive Summary

This report documents the full-universe Indian equity screener: how stocks are classified, fetched, scored, interpreted, and matched to investor risk profiles. The system screens approximately 181 NSE-listed names across eight curated sector buckets using a two-tier analysis model (deep for Banking and IT; standard for all other sectors).

Key design decisions

- Sectors are assigned via universe.yaml, not yfinance sector/industry fields.
- yfinance provides prices, fundamentals, and financial statements; banking asset-quality ratios come from curated JSON.
- Composite scores are peer-relative within cohorts (e.g. private banks vs PSU banks).
- Recommendations use three axes: absolute quality (A-F), peer rank band, and action label.
- Large-cap floor prevents mega-caps from AVOID/SELL unless hard gates fail.
- Rule-based inference (YAML + Python); no LLM in the scoring path.

Current recommendation distribution (typical full screen)

- STRONG BUY / BUY: ~30-35% of universe
- HOLD: ~40-45%
- AVOID / SELL: ~20-25%
- 181/181 tickers typically pass QC after cache refresh

2. System Architecture Overview

Data flows: Configuration (universe.yaml, weights, risk YAML) -> Fetcher (yfinance + cache + NSE fallback) -> QC filter -> Scoring (composite.py / generic.py) -> Interpretation (stock_analyst.py) -> Profile matcher (risk_matcher.py) -> API / CLI / React UI.

Layer	Primary modules	Output
Config	config_loader.py, universe.yaml	Ticker metadata, peer sets
Data	fetcher.py, banking.py, normalize.py	StockMetrics per ticker
Score	composite.py, generic.py, action_matrix.	ScoreResult
Interpret	stock_analyst.py, signals.py	StockInterpretation
Match	risk_matcher.py, questionnaire.py	FitResult picks
Serve	api/main.py, cli.py, web/	REST + UI

3. Sector Taxonomy & Peer Groups

Bucket	Model	Depth	Cyclical	Peer group
banking	BANKING	deep	No	private OR psu sub-cohort
it	IT	deep	No	all IT tickers (~31)
fmcg	FMCG	standard	No	FMCG bucket
pharma	PHARMA	standard	No	Pharma bucket
auto	AUTO	standard	Yes	Auto bucket
energy	ENERGY	standard	Yes	Energy bucket
metals	METALS	standard	Yes	Metals bucket
capital_goods	CAPITAL_GOODS	standard	Yes	Industrials bucket

Banking percentiles are never computed across private and PSU banks together. IT percentiles use the full IT universe as one peer set. Standard sectors rank within their bucket only.

4. Data Acquisition Layer

Primary source: yfinance

- get_info(): PE, PB, ROE, margins, market cap, beta proxies
- history(5y): returns, volatility, drawdown, beta vs Nifty (^NSEI)
- Financial statements: CAGR, credit growth (banks), margin trends

Overrides & fallbacks

- banking_metrics.json: GNPA, NNPA, CAR, NIM (curated, overrides Yahoo)
- ticker_aliases.yaml: symbol fallbacks (M&M.NS, CUB.NS)
- NSE chart API when yfinance history too short
- 24-hour disk cache at .cache/screener/
- sanitize_metrics(): clamp ROE, PE, debt/equity outliers

5. Pipeline & Quality Control

Orchestrator: screener/pipeline.py (ScreenState singleton for API).

QC drop conditions

- fetch_failed = true
- sector_focus = unknown
- price_history_rows < min_price_history_rows (default 20)
- both PE and PB missing
- data_completeness < min_completeness (default 0.35)

After QC: enrich_sector_relative_momentum() sets rs_vs_sector_pct = 6m return minus sector median 6m return.

6. Scoring Framework (Shared Mechanics)

Routing (score_universe)

for each peer group from peer_set(): if banking -> _score_banking(); elif it -> _score_it(); else -> score_generic_group().

Factor utilities (factors.py)

- soft_score_higher/lower: map metric to 0-1 via good/bad thresholds
- percentile_rank: 0-100 within peers, winsorized at 5th/95th percentile
- shrink_percentile: blend toward 50 when peer_count < min_peers_for_rank (10)
- weighted_mean: composite factor blend, skips missing factors

ScoreResult fields

- composite_score (0-100), fundamental_strength (0-1)
- valuation_label: Under | Fair | Over | Unknown
- composite_percentile, peer_rank, peer_count
- quality_grade (A-F), peer_band, recommendation
- red_flag, hard_gate_fail, risk_flags, confidence

7. Sector-Specific Scoring Models

7.1 Banking (deep)

Factor	Weight	Key inputs
Asset quality	30%	GNPA, NNPA, CAR, GNPA peer pctile
Franchise	25%	NIM, ROA, ROE, peer pctiles
Valuation	25%	P/B-ROE residual regression
Momentum	10%	6m return pctile or RS
Risk penalty	10%	Max drawdown pctile

Hard gates: GNPA>=3.5%, NNPA>=1.2%, CAR<12%. Valuation via pb_roe_residual() in banking_valuation.py. Seven risk questions (BQ1-BQ7).

7.2 IT (deep)

Factor	Weight	Key inputs
Margin quality	25%	Op margin, margin trend, ROE
Growth	30%	Rev CAGR (fallback), profit growth
Valuation	25%	PEG, P/E vs peers, FCF yield
Momentum	10%	6m return or RS
Risk penalty	10%	Drawdown pctile

Hard gates: op margin<0, ROE<0, D/E>2.5. Seven risk questions (IQ1-IQ7). Growth fallbacks in growth.py when 3y CAGR missing.

7.3 Standard sectors (generic.py + sector_weights.yaml)

Sector	Quality	Growth	Value	Momentum	Extra
FMCG	35%	15%	25%	10%	risk 15%
PHARMA	35%	25%	20%	10%	risk 10%
AUTO	25%	25%	25%	15%	cyclical
ENERGY	28%	12%	28%	12%	cashflow 18%
METALS	22%	18%	32%	15%	value-heavy
CAPITAL_GOODS	30%	30%	20%	10%	growth focus

Valuation: valuation.py multi-metric vote (P/E, P/B, EV/EBITDA z-scores). Cyclical margin-peak penalty prevents false cheap labels. Five generic risk questions with sector_risk_overrides.yaml bands.

8. Recommendation Engine (Three-Axis v3)

Axis 1: Absolute quality (quality_grade.py)

quality_score = 0.45*fundamental_strength + 0.40*(composite/100) + 0.15*completeness. Grades: A>=0.75, B>=0.55, C>=0.40, D below, F if hard_gate_fail.

Axis 2: Peer band

Band	Percentile
Top	>=70
Upper-Mid	50-70
Lower-Mid	30-50
Bottom	<30

Axis 3: Action label (action_matrix.py)

- Hard fail + Over -> SELL; hard fail else -> AVOID
- Grade A/B + Top/Upper-Mid + not Over -> STRONG BUY or BUY
- Grade A/B + Under -> BUY (value)
- Grade A/B/C + not Bottom -> HOLD
- Bottom + Over -> AVOID
- Large-cap floor (>=500B INR): minimum HOLD unless hard_gate or red_flag
- Confidence <0.35: downgrade one step on action ladder

9. Interpretation & Risk Question Engine

For each stock, stock_analyst.py loads YAML questions, evaluates good/warn/bad signals via signals.py, computes stock_risk_score (0-100), updates confidence, re-runs assign_action(), and generates narrative headline/bull/bear/key_risk.

Analysis type	Sectors	Questions
Deep	Banking, IT	7 sector-specific
Standard	All others	5 generic + overrides

10. Profile Matching & Personalization

User questionnaire maps to RiskProfile (conservative/moderate/growth/aggressive). match_stock() computes profile fit score and exclusion flags.

Profile fit formula

fit = 0.27*risk_alignment + 0.25*quality_norm + 0.18*valuation_fit + 0.10*profile_bonus + 0.20*composite_score

Diversification (risk_profile_matrix.yaml)

- Top 20 picks total, max 3 per sector
- deep_sector_min: at least 2 banking + 2 IT in diversified picks
- Conservative: excludes cyclical sectors, high beta, overvaluation

11. Configuration Reference

File	Purpose
universe.yaml	Ticker lists, buckets, depth, cyclical
settings.yaml	Cache, QC, composite weights, cap tiers
sector_weights.yaml	Standard sector factor weights
banking_metrics.json	Curated bank GNPA/NNPA/CAR/NIM
*_risk_questions.yaml	Banking/IT/generic risk Q definitions
sector_risk_overrides.yaml	Per-sector signal band overrides
risk_profile_matrix.yaml	Matcher penalties, diversification
user_questionnaire.yaml	Investor profile questionnaire
ticker_aliases.yaml	yfinance symbol fallbacks

12. API, UI & Operational Scripts

Endpoint / Script	Purpose
GET /api/screen	Full scored universe
GET /api/stock/{t}/interpret	Stock interpretation
POST /api/recommend	Profile-matched picks
scripts/e2e_audit.py	End-to-end audit
scripts/sector_audit.py	Per-sector breakdown
scripts/rec_audit.py	Blue-chip sanity check
python -m pytest	41+ automated tests

13. Worked Examples

ZYDUSLIFE.NS (Pharma)

Bucket pharma from config. Generic PHARMA weights -> composite ~83. Top peer percentile -> grade A -> STRONG BUY. Stock risk ~23. High profile fit for Capital Preservation.

TCS.NS (IT)

Deep IT scorer. Strong fundamentals but may rank Bottom vs 31 IT peers on composite percentile. Quality grade B. Large-cap floor -> HOLD minimum (not AVOID). Seven IT risk questions including PEG valuation.

HDFCBANK.NS (Banking)

Deep banking scorer within private bank cohort. Curated NNPA/CAR from JSON. P/B-ROE residual valuation. Typically STRONG BUY or BUY with grade A.

Appendix A. Formulas & Thresholds

Banking hard gates

GNPA >= 3.5% | NNPA >= 1.2% | CAR < 12%

IT hard gates

Operating margin < 0 | ROE < 0 | Debt/Equity > 2.5

Generic hard gates

ROE < -5% | Op margin < -2% | D/E > 3 (4 for FMCG/pharma) | Interest coverage < 0.8

Cap tiers (settings.yaml)

Mega: >= INR 2T | Large: >= INR 500B | Floor: HOLD unless hard_gate_fail or red_flag

Appendix B. Glossary

sector_focus	Internal bucket id: banking, it, fmcg, etc.
model_sector	Scoring model key: BANKING, IT, FMCG, ...
analysis_depth	deep (7 Qs) or standard (5 Qs)
composite_score	0-100 weighted factor score within scoring engine
composite_percentile	Rank within peer cohort, 0-100
quality_grade	A-F absolute quality, independent of peer rank
peer_band	Top | Upper-Mid | Lower-Mid | Bottom
action_label	STRONG BUY | BUY | HOLD | AVOID | SELL
fit_score	0-100 profile match for personalized picks
stock_risk_score	0-100 from risk question signals
valuation_label	Under | Fair | Over from peer-relative valuation
rs_vs_sector_pct	6m return minus sector median 6m return